

## How does the Internet work?

To understand how to protect oneself, one must understand how the Internet actually works. What happens when you click on that link in your browser?

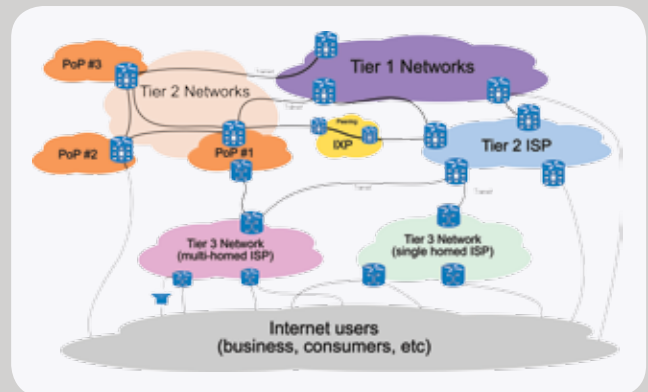
For something as seemingly large and complex as the Internet, it's also surprisingly basic and straight-forward in its operation. After oversimplifying the process considerably, here's a rough overview of the Internet.

Here are some basics that you need to be aware of:

1. **End-points:** The data that you need to access is stored on a PC that is called a server. This could be the place where all the data on [www.thinkdigit.com](http://www.thinkdigit.com) is hosted for example. The device with which you'll access the data from the server is the client. This could be your smartphone, PC or laptop or any other Internet ready device. Both these devices are end-points. The ends of the chain that make up the Internet.
2. **IP Address:** Every end-point / server will use a network card to access the internet and a unique address that's called the IP address. Think of the network card as your cell-phone and the IP-address as your phone number. The IP address itself is not your identity in the same way that a mobile number is not you.
3. **Packets:** All data over the Internet is sent as "packets" of data. Think of it this way, when you click on an image, the data that makes up that image is broken up like a puzzle. Each piece is told where it needs to go and when it gets there, where it needs to fit. When the packet arrives at the destination, it'll find its spot and settle there, when the rest of the packets arrive, the puzzle reassembles itself to give you the image that you requested.
4. **DNS:** Trying to remember phone numbers is hard enough, remembering an IP address even worse. The Domain Name system was put in place to sort out exactly this problem. It's essentially a contact book that keeps track of IP addresses and their associated names. So, [www.google.com](http://www.google.com) can be 74.125.239.96 for example. DNS is a Domain Name Server, a server that hosts a database of domain names that can be referred to. Whenever a new Internet connection is setup, the address of a DNS server is also included.

Let's start with a simple address.

1. You open Google Chrome and type [www.thinkdigit.com](http://www.thinkdigit.com) in the address bar.



2. Your PC translates that data into a request that is sent via a router to your local ISP, which will try to contact a DNS server to figure out the IP address of [www.thinkdigit.com](http://www.thinkdigit.com).
3. The local ISP will have a regional ISP above it. The regional ISP will in-turn contact a Network Service Provider (NSP) that will in-turn contact a Network Access Point (NAP) or IX (Internet exchange). All these ISPs, NSPs and IXs are basically routers. Once the DNS tells the system the IP address that you're looking for, all these routers will try to find that IP address. Each router refers to a routing table, a chart that lists the IP address that the router is responsible for and a default route to the next router in the hierarchy if the required IP address is not listed.
4. Once the addresses are resolved and the server located, your request is finally submitted and the server responds with data. That data is sent in the form of packets (read in detail above). What's interesting here is that the packets don't return to your client via the same route that your request arrived at the server. These packets will discretely pick and choose the most optimal way to arrive at your client and will almost always arrive via multiple routes. This system ensures that the Internet pathways don't get clogged and that a balanced data flow is maintained.
5. That's it! The packets arrive at your client. They're already tagged with the information required to reassemble them and your browser automatically collates the data and interprets it. All this magic happens in a matter of milliseconds.

address, where the request came from and where it was sent. The only way to protect your anonymity online is by making sure that those logs are not kept. VPNs that claim to be secure do exactly that, they delete logs.

If someone wants to find the source of a request that they want to trace, they'll be able to trace it as far as the VPN provider, but no further. Use multiple VPNs and you further enhance your security.

VPNs are also an excellent way to bypass censorship and also to access con-

tent on region-restricted sites including iPlayer, Netflix and Hulu. Do remember that you might very well be breaking the law here and you're on your own if you do take such a risk.

Therefore, to truly take advantage of a VPN you need to access a VPN provider that keeps no records. There is no point in taking steps to protect your privacy if your VPN logs your every move and then sells it off to the highest bidder is there? Another problem is that many of the good VPNs require you to pay a certain sum

of money for their services. Of course, if you can put a price on your privacy, that's where you start.

Proxy servers are very similar but they're not the same thing. Proxy servers only act as intermediaries between you and the rest of the Internet. VPNs act as intermediaries and encrypt your traffic at the same time. Do note that just because a VPN or proxy server is popular doesn't mean that it's good. Consider <http://hidemyass.com/>; it was recently accused